

Lab 3 – Written Architecture Justification

Multi-Vendor Artisan E-Commerce Marketplace

Selected Architecture: Layered Architecture

Architecture Selection

We chose Layered Architecture for the Multi-Vendor Artisan E-Commerce Marketplace. The system is organized into Presentation, Business, and Data layers. The Marketplace User Interface communicates with the business components, while the business components access the Marketplace Database.

Reason 1 – Clear Separation of Responsibilities

Layered Architecture provides a clear separation of concerns. The Presentation Layer handles shopper and artisan-vendor interactions. The Business Layer manages vendor storefronts, product catalogues, shopping carts, orders, payments, and the 5% platform commission and vendor-payout calculation. The Data Layer manages persistent marketplace data. This makes the system easier to understand, maintain, and modify.

Reason 2 – Suitable for the Marketplace Workflow

The marketplace follows a structured flow from user interaction to business processing and then data storage. Orders and payments require coordinated business logic, including payment verification, order creation, and splitting the customer's payment among vendors after deducting the 5% platform commission. Keeping this logic in the Business Layer provides a simple and organized design suitable for the system.

Security Advantage

Layered Architecture separates sensitive business and data operations from the user interface. Payment processing, vendor payout calculation, commission handling, and order information remain within the Business Layer instead of being directly exposed to the Presentation Layer. Database access is also controlled through the Business Layer.

Performance Benefit

The Product Catalog Manager can optimize catalogue retrieval to support the requirement that catalogue loading remain below 500 ms under simulated peak load. High-resolution product images can use CDN caching to reduce repeated retrieval and improve response time. Layered Architecture supports this separation, although the architecture itself does not guarantee the 500 ms target.